



Problem Set 7

Due: 8 Jul 2024, 3:30 p.m.

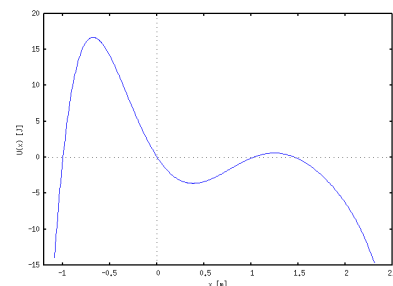
Problem 1. Given the potential energy $U(x, y) = xy^2 + yx^2$, (a) find the corresponding force field and visualize it (use a computer); (b) on the same graph sketch a few equipotential lines, i.e. lines defined by equation $U(x, y) = U_0$, for different values of U_0 ; (c) comment on the graph; (d) calculate work done by this force on a particle moving from (0,0) to (1,1) along a straight line; (e) same as (d) but along the parabola $y = x^2$.

Note. In (d) and (e) use the simplest possible method.

(3/2 + 1 + 1 + 1 + 1 points)

Problem 2. The figure shows the graph of the potential energy function $U = U(x)$ for a particle moving along the x axis in some force field. Identify all the equilibrium positions and tell which of them are stable/unstable.

(3/2 points)



Problem 3. The Lennard–Jones potential energy $U = U_0 \left[\left(\frac{R_0}{r} \right)^{12} - 2 \left(\frac{R_0}{r} \right)^6 \right]$, also called the L–J potential or 6–12 potential, is often used to approximate the potential energy associated with interaction between a pair of neutral atoms or molecules separated by distance r .

- Find the corresponding force (it is the force exerted by one atom/molecule on the other). Sketch the graphs of both the potential energy and the force as functions of r . Which term in the force is responsible for attraction and which for repulsion?
- What is the interpretation of the parameters U_0 and R_0 (both are positive)?
- Introducing a new variable $x = r - R_0$, find an approximate expression for the force in the regime $|x/R_0| \ll 1$. Interpret your result in terms of oscillations. Find their period.

(3/2 + 1/2 + 3 points)

Problem 4. What is the period of *small* oscillations about the stable equilibrium position in the potential field with the potential energy $U(x) = U_0 \tan^2 \alpha x$, where U_0 and α are positive constants (what are their units)? The mass of the oscillating particle is m .

(2 points)

Problem 5. One end of a spring with the spring constant k is attached to the wall and the other to a block with mass $2m$ resting on a smooth surface. Initially, the spring is neither stretched nor compressed. A ball with mass m , moving with speed v_0 , hits the block in an elastic collision. What is the maximum compression of the spring after the collision?

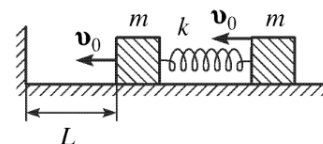
(4 points)



Problem 6. Two identical blocks with mass m are connected by a light spring with spring constant k . Initially, the spring is neither compressed nor stretched and the system slides with speed v_0 on a frictionless surface towards a wall (see figure below). At $t = 0$ the left block is at distance L from the wall. Find the time after which the center of mass of this system will be again at the position it was when $t = 0$.

Assume that the block collides with the wall elastically, and the duration of the collision is negligibly short.

(4 points)



Problem 7. Consider a system of three particles with masses $m_1 = 5$ g, $m_2 = 10$ g, $m_3 = 15$ g. At the initial instant of time $t = 0$ the particles were placed at points (3,4,5), (2,4,6), and (0,0,0), respectively (the coordinates are given in centimeters). The particles are acted upon by external forces whose sum is a vector of magnitude $F = 0.05$ N pointing along the x axis, in the positive x direction.

Find the position of the center of mass of the system for $t = 2$ s.

(2 points)

Problem 8. Consider a rod of length l , mass m , and cross-sectional area A . Let us set the origin of the coordinate system at one end of the rod with the positive x -axis along the symmetry axis of the rod. (a) If the bulk density $\rho = m/Al$ of the rod is constant, show that the x -coordinate of the center of mass of the rod is at its half of its length. (b) If the density of the rod varies linearly with x — that is, $\rho = \alpha x$, where α is a positive constant — calculate the x -coordinate of the rod's center of mass.

(1 + 2 points)

Problem 9. A fisherman (with mass m) walks from one end of his boat of mass M and length l to the other. Find the distance the boat has moved with respect to the still surface of a lake the boat is on.

(2 points)

Problem 10. A toy-rocket with initial mass m_0 is placed at rest on a frictionless horizontal surface at some distance from a wall. At $t = 0$, the engine is turned on and the rocket starts moving in the direction perpendicular to the wall colliding with it after time T . The collision is elastic and the rocket does not change its orientation with respect to the wall. How much time after the collision does it take until the (instantaneous) velocity of the rocket is zero?

Assume that the burned fuel is ejected from the rocket at a constant speed (with respect to the rocket) and the mass of the rocket decreases with time at a constant rate α .

What is the numerical answer if $m = 1$ kg, $\alpha = 0.01$ kg/s, and $T = 10$ s?

(5 points)

